Verification Report for CPU Scheduler

1. *Objective*

The purpose of this report is to validate that the simulator produces correct outputs by cross-checking its results against manually computed execution traces. Verification specifically targets the backend scheduling logic, confirming that algorithms accurately compute. This ensures the application is not only visually functional but also computationally accurate and reliable.

2. *Formula*

TAT = CT – AT WT = TAT – BT

Avg. CT or TAT or WT = (∑CT or ∑TAT or ∑WT) / number of processes

3. *Sample Data*

Processes	Arrival Time	CPU Burst Time
P1	1	7
P2	4	2
P3	4	1
P4	7	6
P5	10	4

AT = Actual Time CT = Completion Time WT = Waiting Time

BT = CPU Burst Time TAT = TurnAround Time

FCFS (First Come First Serve)

Processes	Arrival Time	Burst Time	CT	TAT	WT
P1	1	7	8	7	0
P2	4	2	10	6	4
P3	4	1	11	7	6
P4	7	6	17	10	4
P5	10	4	21	11	7
Total = 5	26	20	13.4 (Avg. CT)	8.2 (Avg. TAT)	4.2 (Avg. WT)

Ready Queue: ~~P5~~ ~~P4~~ ~~P3~~ ~~P2~~ ~~P1~~ ←

Gantt Chart

Idle	P_1	P_2	P_3	P_4	P_5

0 1 8 10 11 17 21

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RR (Round Robin) – Preemptive

Suppose Time Quantum (Time Slice) = 2

Processes	Arrival Time	Burst Time	CT	TAT	WT
P1	1	7 ~~5~~ ~~3~~ ~~1~~ 0	15	14	7
P2	4	2	7	3	1
P3	4	1	8	4	3
P4	7	6 ~~4~~ ~~2~~ 0	21	14	8
P5	10	4 ~~2~~ 0	19	9	5
Total = 5	26	20	14 (Avg. CT)	8.8 (Avg. TAT)	4.8 (Avg. WT)

Ready Queue: ~~P4~~ ~~P5~~ ~~P4~~ ~~P1~~ ~~P5~~ ~~P4~~ ~~P1~~ ~~P3~~ ~~P2~~ ~~P1~~ ~~P1~~ ←

Gantt Chart

Idle	P_1	P_1	P_2	P_3	P_1	P_4	P_5	P_1	P_4	P_5	P_4

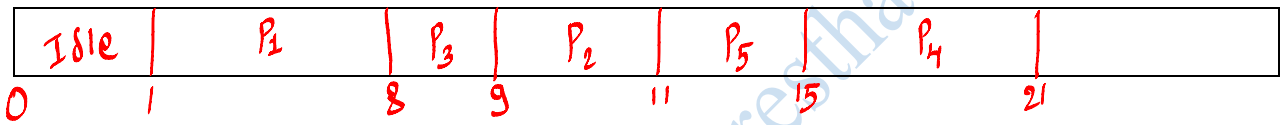
0 1 3 5 7 8 10 12 14 15 17 19 21

SJF (Shortest Job First) - Non-Preemptive

Processes	Arrival Time	Burst Time	CT	TAT	WT
P1	1	7	8	7	0
P2	4	2	11	7	5
P3	4	1	9	5	4
P4	7	6	21	14	8
P5	10	4	15	5	1
Total = 5	26	20	12.8 (Avg. CT)	7.6 (Avg. TAT)	3.6 (Avg. WT)

Ready Queue: P4 P5 P2 P3 P1

Gantt Chart



{ SJF = Once it starts, it finishes it completely (no-interruption)
SRTF = If a shorter job arrives, it interrupts the current one. }

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SRTF (Shortest Remaining Time First) - Preemptive

Processes	Arrival Time	Burst Time	CT	TAT	WT
P1	1	7 ~~4~~ ~~1~~ 0	11	10	3
P2	4	2	7	3	1
P3	4	1	5	1	0
P4	7	6	21	14	8
P5	10	4	15	5	1
Total = 5	26	20	11.8 (Avg. CT)	6.6 (Avg. TAT)	2.6 (Avg. WT)

Ready Queue: P4 P5 P1 P1 P2 P3 P1

Gantt Chart

Final Summary

The CPU Scheduler was thoroughly verified using manually computed results for FCFS, Round Robin (Time Quantum = 2), SJF (Non-Preemptive), and SRTF (Preemptive). For each algorithm, Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) were calculated using standard scheduling formulas and validated against the simulator's output. The Gantt charts and ready-queue sequences were also manually constructed to ensure accurate process ordering and preemption behavior.

Across all four algorithms, *the simulator's results matched the manually derived values, confirming that the implementation correctly follows the expected scheduling logic*. Based on this verification, the **CPU Scheduler** is validated as accurate, reliable, and suitable for educational and demonstration purposes.

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